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Multiobjective optimization of strategies for operation and testing of low-demand safety instrumented systems using a genetic algorithm and fault trees

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ABSTRACT

Since low-demand safety instrumented systems (SISs) do not operate continuously, their failures are often only detected when the system is demanded or tested. The conduction of tests, besides adding costs, can raise risks of failure on demand during their execution and also increase the frequency of spurious activation. Additionally, it is often necessary to interrupt production to carry out tests. In light of this scenario, this paper presents a model to optimize strategies for operation and testing of these systems, applying modeling by fault trees associated with optimization by a genetic algorithm. Its main differences are: (i) ability to represent four modes of operation and test them for each SIS subsystem; (ii) ability to represent a SIS that executes more than one safety instrumented function; (iii) ability to keep track of the down-time generated in the production system; and (iv) alteration of a genetic selection mechanism that permits identification of more efficient solutions with smaller influence on the optimization parameters. These aspects are presented by applying this model in three case studies. The results obtained show the applicability of the proposed approach and its potential to help make more informed decisions.

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1. Introduction

Safety instrumented systems (SISs) provide a layer of independent protection. They are used to control or mitigate risks related to the operation of many industrial installations [1]. SISs can be subdivided into three subsystems: (i) sensor elements (SEs); (ii) logic solver (LS) units; and (iii) final elements (FEs). They have two main functions: (a) to activate the final elements when a specific demand occurs in the system being monitored, where failure to carry out this function is called a failure on demand or failure to function (FTF); and (b) to remain deactivated without the occurrence of a defined demand, where failure is called spurious activation, or a spurious trip (ST) [2].

Each SIS is responsible for executing one or more safety functions. For each of these there is a safety instrumented function (*SIF*) and an associated safety integrity level (SIL) [3]. There are three operating modes of a *SIF* [4]: (i) low demand, when the function is required to operate less often than once per year; (ii) high demand, when the function is required to operate more than once a year; and (iii) continuous, when the function has to assure safety of the system during all normal operations. The parameter to assess the reliability of a low-demand *SIF*, with respect to the FTF, is the probability of failure on demand (*PFD*), while for high-demand and continuous modes the parameter is the probability of failure per hour (PFH) [4]. Finally, the parameter to assess reliability regarding ST is the spurious trip rate (STR) [5].

In a SIS operating under low demand, some failures to their components can remain hidden, only being detected when the system is tested or a real demand occurs [2]. Although periodic tests can detect some hidden problems, they increase costs and can heighten risks during their execution, by increasing the *PFD* during testing [6], also increasing the spurious trip rate (STR) [7]. In factories where production is continuous, it is normally necessary to interrupt production of the protected system to test some SIS components, thus causing a conflict between ongoing production and execution of the test [8]. On the other hand, various tests can be conducted during scheduled shutdowns for maintenance, which normally happen at intervals ranging from two to six years. In these situations, it is possible to perform tests without increasing the likelihood of spurious activation or impairing production.

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Many studies have investigated questions related to estimating the reliability parameters of safety instrumented systems. However, these studies only considered part of the subsystems, did not estimate the rate of spurious activation and disregarded the possible strategies for operation and execution of the tests [9–11]. In turn, [12] estimated the spurious activation but disregarded the possible strategies for operation and test execution. Other studies [6,13] considered spurious activation and two testing modes in their SIS test optimization models. However, in some situations the selection between the testing modes provided solutions that cannot be applied in real systems in the same way they were modeled. These studies also disregarded, in estimating the total operating cost of these systems, the fraction caused by the production down-time during execution of the tests or repairs to the safety system. The PDS method [5] considers three operating and testing philosophies and evaluates the down-time of the productive system, using simplified equations for this purpose. Nevertheless, this method does not consider the costs of its formulation, nor does it handle optimization of its tests. Besides this, more than one safety function can be implemented in a single SIS, through sharing of some of its subsystems [14]. In these situations, the operating and testing strategy applied must meet the needs of the various safety functions in correlated form.

Therefore, the main goal of this paper is to present a model for multiobjective optimization of the strategies for operation and testing of low-demand SISs. The model considers the probability of failure of each of the instrumented functions, to try to satisfy the required SIL and minimize the impact on the cost over the installation's life cycle. Based on these objectives, it was possible to formulate more realistic models to represent the interrelationship of one or more safety functions with the production system. It was also possible to identify workable testing strategies, even in complex systems, able to assure the integrity of the safety system while minimizing its impact on the business.

The main advantages of the approach presented are: (i) the model's ability to represent four different modes of operation and to test each subsystem of the SIS, allowing selection of the most suitable mode for each context by means of optimization; (ii) the ability to represent a SIS that carries out more than one instrumented function, with or without sharing of components; (iii) the ability to keep track of the production down-time generated by SIS tests and repairs; and (iv) the possibility of altering a selection mechanism applied to the genetic algorithm, enabling identification of more efficient solutions with less influence of the optimization parameters.

2. Probabilistic model

To compare the safety integrity level (SIL) and composition of the associated costs over the entire life cycle of each *SIF*, we calculated the parameters PFD_{avg}, STR_{avg} and U_{avg}, where the last one represents the average unavailability of the protected system resulting from maintenance or testing of the *SIF*.

In developing the model we considered four possible modes of operation and testing, applicable to each subsystem of the SIS. These modes were defined based on the philosophies for operation and testing presented in [5] and the testing modes utilized in [15]. They are described below

i. *M*1—always stop: production will be interrupted any time a failure is detected or a test is conducted;
ii. *M*2—always open: the voting logic will be altered any time a failure occurs or components are tested. This change in voting logic means the tested or failed component votes for activation of the *SIF*.
iii. *M*3—always close: the voting logic will be altered any time a failure occurs or components are tested. However, in this case the change in the voting logic means the tested or failed component cannot vote to activate the *SIF*.
iv. *M*4—always bypass: the *SIF* will be removed from operation any time a failure is detected or a test is conducted, with the system operating without the referred protection during the period necessary to restore the normal condition.

Modes *M*1 and *M*4 are only applicable to subsystems where the tests are conducted of all the components simultaneously.

We parameterized the execution of the complete tests using three variables for each SIS subsystem: (i) execution time of the first test (T_{FT}); (ii) testing interval factor (f_{TI}), with values in the interval [0, 1] and; (iii) testing cycle period (T_C), which represents the time between conducting a test of any of the components and the next test of the same component. If $f_{TI}=0$, the tests of the various components are carried out simultaneously and if $f_{TI}=1$ the tests are evenly spaced in the interval T_C. These definitions are similar to those used in the SIS testing optimization study of [6]. For the subsystems of final elements, we added one more variable, with the aim of representing the partial movement or partial stroke tests (PSTs) [16]. This variable, N_{PST}, represents the number of partial tests that will be conducted in each complete testing cycle (T_C).

The complete tests have defined duration (T_D) and the partial tests (PST) are considered to be instantaneous. If the mode of operation chosen is *M*1, we assume the test duration is the mean time to stop and start the protected system (*MTTSS*). Besides this, the proposed model considers the existence of testing windows, resulting from scheduled shutdowns for maintenance of the production system, in a defined period (t_{max}).

We consider homogeneous components in each subsystem and that the time to failure of each of these obeys the homogeneous Poisson process. The failure rate of each component is subdivided into the four fractions: dangerous detected (λ_{DD}), dangerous undetected (λ_{DU}), safe detected (λ_{SD}) and safe undetected (λ_{SU}) [2,4]. The model presented here also considers a fraction of failures that is independent of conduction of the tests (P_{TIF}) [5]. Besides this, the PST can detect a fraction θ_{PST} of the undetected failures [17]. The mean time to repair (*MTTR*) and mean time to stop and start (*MTTSS*) of the protected system are considered to have constant duration. The model used to represent the failures with common cause is the modified beta factor model [5], which can easily be reduced to the beta factor model [18] and is described in the IEC standard [4].

The values of the basic events of the fault trees were obtained from the equations presented. The value of the top event was computed discretely each hour in the interval [1, t_{max}]. Its average values were obtained from the mean of these top values, similar to that presented in [6]. The equations of the probability of occurrence of the basic events considered three distinct moments: (i) normal operation—when the component is not subject to testing or repair; (ii) during testing—when the component is being tested; and (iii) during repair—period after a test, when a repair might being carried out to correct a failure identified.

2.1. Probability of failure on demand

To obtain the *PFD* of each *SIF*, we used the quantitative fault tree technique and its complete structure functions, since its minimum cutsets can be changed while conducting the tests [15].

Fig. 1 shows the fault tree to obtain the *PFD*, where the top event is "*SIF* Unavailable". In the model developed here, the top event occurs any time one of the *SIF* subsystems is unable to operate ("*SIF* Failure"), or any time one of the subsystems is

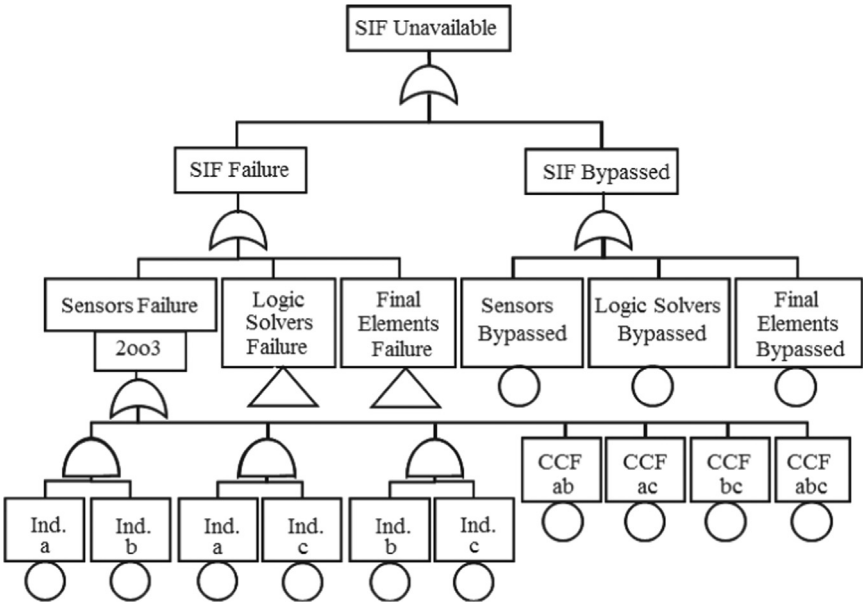


Fig. 1. Fault tree for unavailability of the *SIF*. Ind a: independent failure of component a, CCF ab: common cause failure of components a and b.

bypassed to perform a test or repair ("*SIF* Bypassed"). The branch called "*SIF* Failure" contains the developed fault tree for sensor subsystem, which makes use of a typical fault tree for subsystems with *2oo3* voting logic, obtained from [6].

The probabilities that the sensors, logic solvers or final elements are being bypassed will always be equal to zero for all operation and testing modes except for the subsystems that use the *M*4 mode. In this case, the probability of the referred subsystem being bypassed is given by Eq. (1), for normal operation periods, i.e., when none of its components is being subjected to tests or repairs.

In Eq. (1), the term $((N-C_N\beta)\lambda_{DD})$ denotes the total frequency of dangerous detected failures in a subsystem with N components, considering the independent and common cause failures. The term $(\left(N-(C_N+C_{(N-M+1)ooN})\beta\right)\lambda_{SD})$ indicates the frequency of safe detected failures which cannot lead to spurious activation in a subsystem with *MooN* voting logic. The values of C_N and C_{MooN} can be obtained from [5]. The multiplication of a frequency by *MTTR* is the result of the following approximation: $(MTTR/(MTTF+MTTR))=(\lambda MTTR/(1+\lambda MTTR))\approx\lambda MTTR$ [2]

$$P(t)=\begin{cases}\lambda_{DD}MTTR, & 1oo1\\ (N-C_N\beta)\lambda_{DD}MTTR, & 1ooN\\ ((N-C_N\beta)\lambda_{DD}+\left(N-(C_N+C_{(N-M+1)ooN})\beta\right)\lambda_{SD})MTTR, & MooN\ \ and\ \ M>1\end{cases} \tag{1}$$

During periods when tests are being conducted, that probability is equal to 1, since when using operation mode *M*4 the *SIF* will always be bypassed and thus unavailable for activation in case of a real demand. In the period after realization of the test, with duration *MTTR*, any component that is identified as not working will have to be repaired. Therefore, in this period we have continuity of the testing condition ($P(t)=1$), with probability of identifying a failure of $1-e^{-N(\lambda_{DU}+\lambda_{SU})T}$ or complementary probability ($e^{-N(\lambda_{DU}+\lambda_{SU})T}$) of being returned to normal condition (Eq. 1). T is the time since the last test of component i and N is the number of components tested simultaneously.

Eqs. (2)–(4) are used to calculate the *PFD* of each of the basic events of a subsystem with *MooN* voting logic of the "*SIF* Failure" branch. Eq. (2) gives the probability of a failure on demand independent of each of the components i (PFD^i) during the normal operation period. If the operation and testing mode selected is *M*3, the expression contains three terms: (i) $(\lambda_{DDI}+\lambda_{SDI})MTTR$, which denotes the unavailability of the component due to the occurrence of dangerous detected and safe detected failures. That term is specific for operation mode *M*3 because the component's voting will be prevented after the occurrence of these detected failures, preventing it from indicating the need to activate the *SIF* during the repair period (*MTTR*). For the other operating modes, this type of event does not increase the PFD^i. The term λ_{SDI} is not considered when $M=1$ since this event leads to spurious activation of the system and hence prevents the failure on demand. (ii) P_{TIFI}, which denotes the independent fraction from the standpoint of the component, i.e., the failures that occur regardless of the conduction of the tests (P_{TIF}) [5]. (iii) $1-e^{-\lambda_{DUI}t^i}$, which accounts for the probability of a failure on demand resulting from dangerous undetected failures (which are only detected when performing the tests), where λ_{DUI} is the rate of occurrence of those failures and t^i is the time since the last test of component i, or for periods that precede the first test, the time since start-up of the protection system [15]

$$PFD^i(t)=\begin{cases}P_{TIFI}+(1-e^{-\lambda_{DUI}t^i}), & M1,\ M2\ \text{or}\ M4\\ \lambda_{DDI}MTTR+P_{TIFI}+(1-e^{-\lambda_{DUI}t^i}), & M3\ \text{and}\ M=1\\ (\lambda_{DDI}+\lambda_{SDI})MTTR+P_{TIFI}+(1-e^{-\lambda_{DUI}t^i}), & M3\ \text{and}\ M>1\end{cases} \tag{2}$$

Eq. (3) gives the probability of a failure on demand of two components i and j with common cause, when both components are operating normally. This expression differs from Eq. (2) by: (i) the alterations in the failure frequencies, which in this case refer to the frequencies due to common cause in components i and j, (ii) by exclusion of the term λ^{ij}_{SDI} in mode *M*3 with $M=2$, since this event leads to spurious activation of the system, and (iii) by substitution of t^i by $min(t^i, t^j)$, which is the time since the last test of component i or component j, because common cause failures of those two components are always detected any time one of them is tested [9]

$$PFD^{ij}(t)=\begin{cases}P^{ij}_{TIFC}+1-e^{-\lambda^{ij}_{DUC}\min(t^i,\,t^j)}, & M1,M2\ \text{or}\ M4\\ \lambda^{ij}_{DDC}MTTR+P^{ij}_{TIFC}+1-e^{-\lambda^{ij}_{DUC}\min(t^i,\,t^j)}, & M3\ \text{and}\ M\le 2\\ (\lambda^{ij}_{DDC}+\lambda^{ij}_{SDC})MTTR+P^{ij}_{TIFC}+1-e^{-\lambda^{ij}_{DUC}\min(t^i,\,t^j)}, & M3\ \text{and}\ M>2\end{cases} \tag{3}$$

To obtain the probability of failure on demand with common cause of components *i*, *j* and *k*, when all three are operating normally, the same logic is used as in Eq. (3).

Eq. (4) presents the *PFD* of component *i* when it is being tested. If modes *M*1 or *M*2 are used, PFD^i is equal to zero during the test, since: (i) in mode *M*1 the operation of the protected system will be interrupted to conduct the test, so the system will not be subject to a failure on demand; and (ii) in mode *M*2 the component will be under safe condition during the test (voting for activation of the *SIF*). In turn, when modes *M*3 of *M*4 are selected, the ability of the component to demand activation of the *SIF* during the test is disabled, so that in these cases PFD^i equals 1

$$PFD^i(t) = \begin{cases} 0, & M1 \text{ or } M2 \\ 1, & M3 \text{ or } M4 \end{cases} \quad (4)$$

The probability of a failure on demand due to common cause failure of any two components *i* and *j* during the interval when component *i* is being tested is nil for modes *M*1, *M*2 and *M*4. The reasons are that: (i) in mode *M*1, the production will be interrupted, so a failure on demand makes no sense; (ii) in mode *M*2, the component will be tested while open, i.e., while voting for activation of the *SIF*, preventing common cause failure of components *i* and *j*; and (iii) in mode *M*4, the *SIF* will be bypassed, so it will be out of operation. For mode *M*3, the probability of failure on demand due to common cause failure of any two components is still possible and continues in relation to the period before the test (Eq. 3), since only at the end of testing component *i* can the possibility of common cause failure of both components be discarded.

The PFD^i during the period when the component can be undergoing repair, with duration of *MTTR*, is obtained by considering the probability of extension of the condition of Eq. (4), with probability of $1-e^{-N(\lambda_{DU}+\lambda_{SU})T}$, associated with the return to the condition given by Eq. (2), with complementary likelihood $(e^{-N(\lambda_{DU}+\lambda_{SU})T})$.

2.2. Spurious trip rate

To obtain the *STR*, just as for the *PFD*, we used the quantitative fault tree technique. However, since in this case the expected response in the top event is in terms of frequency rather than probability, instead of the structure function we used the rule of the sum of the frequencies in the "OR" gates ($Fr^{A\ or\ B} = Fr^A + Fr^B$) [6]. The frequency of an "AND" gate is quantified by $Fr^{A\ and\ B} = Fr^A Pr^B + Fr^B Pr^A$, where Fr^i and Pr^i are the frequency and probability of safe fault of basic event *i* [12].

Fig. 2 shows the fault tree to obtain the *STR*, where the top event is spurious activation. In the model developed in this study, the top event occurs any time one of the *SIF* subsystems demands its activation ("Spurious fault"), and the *SIF* is not bypassed ("*SIF* not bypassed"), since when bypassed, the *SIF* cannot be activated. In the event called "Spurious fault", only the branch of the fault tree of the sensor subsystem is developed, making use of a typical fault tree for subsystems with *2oo3* voting logic, as in [6], plus the term for the spurious trip rate induced by the test (Fr_{TIS}). The probability of occurrence of "*SIF* not bypassed" is obtained from the complementary term of the "*SIF* bypassed" branch of the fault tree of the *PFD* presented in Fig. 1.

The parameter Fr_{TIS} is obtained by considering the probability of a spurious fault during testing (P_{TIS}), which represents the occurrence of a human error during the test, and the frequency with which the subsystem is tested.

To represent the subsystem during testing of component *a*, it was necessary to reconfigure part of the fault tree presented in Fig. 2 in cases *M*2 and *M*3. These reconfigurations are depicted in Fig. 3. For cases *M*1 and *M*4, no reconfiguration is necessary, since in the first case the operation of the protected system will be interrupted to perform the test, and in the second case the *SIF* will be bypassed.

The fault tree in Fig. 3, for *M*2, was obtained from [6]. In turn, for *M*3, we took into consideration the impossibility of a safe fault of component *a*, which is being tested. Therefore, the tree branches shown in Fig. 2, which depend on the independent failure of component *a*, were eliminated. The common cause failures of components *a* and *b* and *a* and *c* were likewise eliminated, since their occurrence is impossible. In contrast, the term referring to the common cause failure of components *a*, *b* and *c* was maintained, since this becomes complementary to the common cause term of components *a* and *b*, which leads to spurious activation of the *SIF* subsystem that has *2oo3* configuration.

To calculate the frequencies and probabilities of the basic events, we used Eqs. (5) and (6). Eq. (5) presents the frequency of independent safe failures of component *i*. When applying operation modes *M*1, *M*3 and *M*4, the safe failure frequency is obtained by the independent fraction of the total detected and undetected safe failures ($\lambda_{SUI}+\lambda_{SDI}$). In the case of mode *M*2, when a dangerous failure (λ_{DDI}) is detected, the component immediately votes to activate the *SIF*. Therefore, for *M*2 the total frequency of independent safe failures is given by $\lambda_{SUI}+\lambda_{SDI}+\lambda_{DDI}$

$$Fr^i(t) = \begin{cases} \lambda_{SUI}+\lambda_{SDI}, & M1, M3 \text{ or } M4 \\ \lambda_{SUI}+\lambda_{SDI}+\lambda_{DDI}, & M2 \end{cases} \quad (5)$$

The same logic is employed to obtain $Fr^{ij}(t)$ and $Fr^{ijk}(t)$, which are the frequency of common cause safe failures of components *i* and *j* and the frequency of common cause safe failures of components *i*, *j* and *k*. In these cases, all the failure frequencies of component *i*, from Eq. (5), are replaced by common cause frequencies of components *i* and *j* and *i*, *j* and *k*, respectively. Considering the modified beta factor in a *2oo3* subsystem, the required failure frequencies are obtained by the following expressions: $\lambda^{ij}_{SUC} = 0.7\beta\lambda_{SU}$, $\lambda^{ij}_{SDC} = 0.7\beta\lambda_{SD}$, $\lambda^{ij}_{DDC} = 0.7\beta\lambda_{DD}$, $\lambda^{ijk}_{SUC} = 0.3\beta\lambda_{SU}$, $\lambda^{ijk}_{SDC} = 0.3\beta\lambda_{SD}$ and $\lambda^{ijk}_{DDC} = 0.3\beta\lambda_{DD}$.

Eq. (6) presents the probability of an independent safe failure of component *i* during normal operation. For modes *M*1, *M*3 and *M*4, the probability of safe failure of component *i* at instant *t* is given by $1-e^{-\lambda_{SUI}t^i}$, where λ_{SUI} is the rate of undetected independent safe failures. For mode *M*2, any time failures are detected, the component immediately votes to activate the *SIF*. Therefore, the equation for the other modes additionally contains $(\lambda_{DDI}+\lambda_{SDI})MTTR$, which indicates the unavailability caused by the independent fraction of dangerous and safe detected failures, keeping component *i* unavailable for the repair time (*MTTR*)

$$Pr^i(t) = \begin{cases} 1-e^{-\lambda_{SUI}t^i}, & M1, M3, M4 \\ (\lambda_{DDI}+\lambda_{SDI})MTTR + 1-e^{-\lambda_{SUI}t^i}, & M2 \end{cases} \quad (6)$$

While component *i* is being tested, that probability is 0 for modes *M*1, *M*3 and *M*4, since in these situations the protected system will be shut down, a dangerous condition will occur or the *SIF* will be bypassed, respectively. For *M*2, the component assumes the safe failure condition during the test due to the operation and testing mode applied, so the probability is equal to 1.

The probability during the period when the component might be undergoing repair, with duration equal to *MTTR*, is obtained by considering the probability of extension of the test condition, with probability equal to $1-e^{-N(\lambda_{DU}+\lambda_{SU})T}$, associated with the return to the condition given by Eq. (6), with complementary probability $(e^{-N(\lambda_{DU}+\lambda_{SU})T})$.

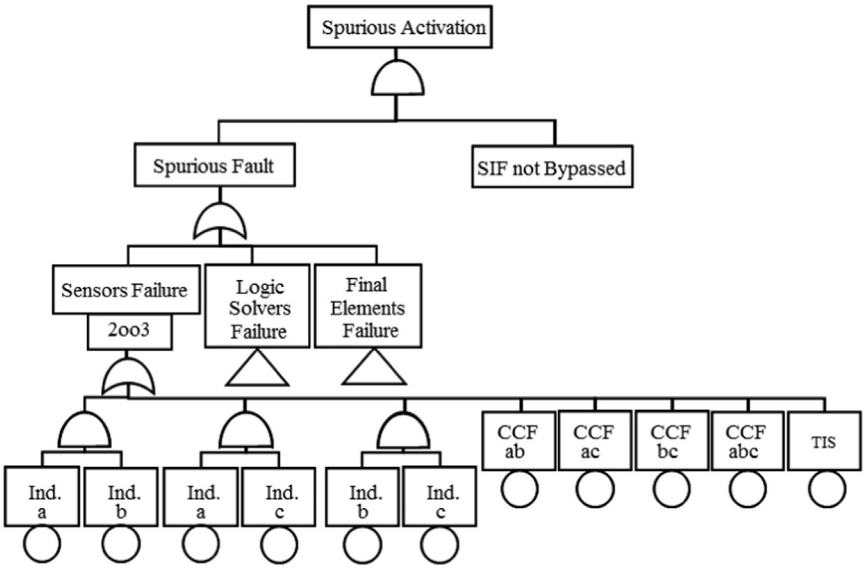


Fig. 2. Fault tree for spurious trip—normal operation.

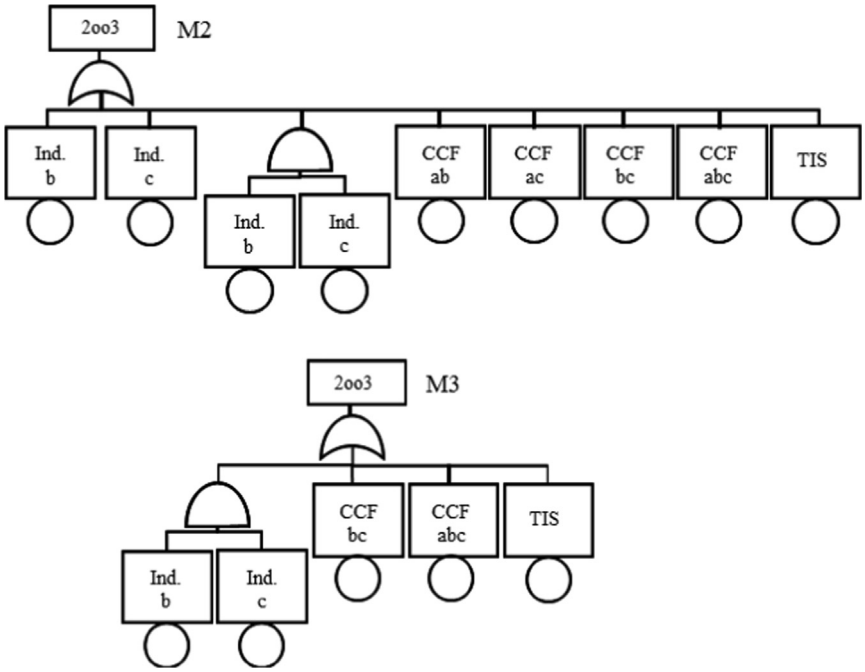


Fig. 3. Fault tree for safe failure of *2oo3* subsystems—during testing.

2.3. Production losses due to testing and maintenance

The unavailability of the protected system because of maintenance and testing of the *SIF* ($U(t)$) was obtained by using the fault tree shown in Fig. 4. The protected system will be unavailable any time the failure condition is reached or a test is conducted that leads to the need for shutdown demanded by the sensors, logic solvers or final elements.

In mode $M1$ there is a possibility of reaching a failure condition that requires the system to be shut down. In the other modes, the system protected by the *SIF* always continues to operate ($P(t)=0$), except in the occurrence of a spurious trip, which is accounted for by the *STR* term.

For subsystems in mode $M1$, the probabilities that the subsystems of sensors, logic solvers or final elements will demand shutdown are given by the same expressions to calculate the "*SIF* bypassed" branch of the fault tree of Fig. 1. Nevertheless, since the

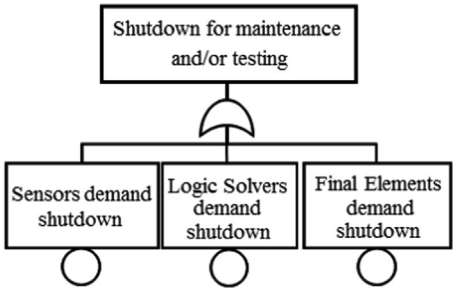


Fig. 4. Fault tree for shutdown for maintenance and/or testing.

duration of interruption of the system is given by *MTTSS*, it is necessary to replace *MTTR* by *MTTSS* in the equations.

2.4. Approach for a SIS with more than one SIF

The approach developed in this paper allows evaluation of the parameters PFD_{avg}, STR_{avg} and U_{avg} for various *SIF*s that share subsystems, as shown in the example presented in Fig. 5 and described in [14]. In this example two *SIF*s use the same set of logic solvers. In this case, the parameters of interest are obtained for each *SIF*, allowing comparison of the SIL required for each of them.

We compared the results obtained from this probabilistic model with those obtained by other authors [5,9–12]. These results are summarized in Table 1. We only observed significant deviations in the comparison with studies [10,11], resulting from different formulations of the failures that occur regardless of the conduction of the tests (P_{TIF}). In this study, this parcel was considered constant in time, while in studies [10,11] it was considered to be an exponential probability of failure.

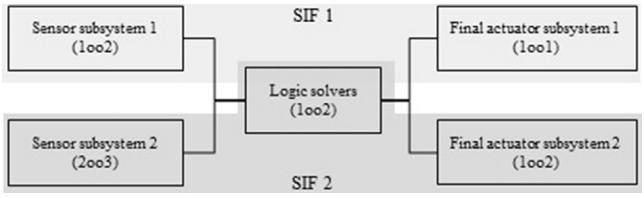


Fig. 5. SIS with two *SIF*s.

3. Multiobjective optimization based on the NSGA-II method

Optimization of the reliability and availability of systems is based on quantifying how the design, testing and maintenance strategies affect reliability, availability, maintainability and costs. Most of these problems can have more than one objective to be optimized, being classified as multi-objective optimization problems [19].

The genetic algorithm is an optimization technique inspired by the evolutionary theory of the origin of species and is a very useful method to assess multi-objective problems. Its application allows identification of a set of optimized solutions with respect to the objectives (Pareto frontier) [12,20]. The basics for application of genetic algorithms to address reliability, availability, maintainability and safety applications have been described by [19].

To optimize the strategies of operation and testing of low-demand safety instrumented systems, we inserted the probabilistic model described in a multiobjective evolutionary optimization algorithm called NSGA-II, which uses elitism, dominance and preservation of diversity [21].

The genetic code defined to represent each operation and testing strategy contains two sets of variables for each SIS subsystem: (i) binary: four positions, where the first two define the operation and testing mode, the third defines whether the tests will be carried out, and the fourth whether the tests will be performed jointly on the components; and (ii) continuous: three positions, which contain T_{FT}, T_C, and f_{TI}. Another binary position is added in subsystems of final elements, to define whether the PST will be executed, along with a continuous position, which contains the value of N_{PST}. The crossover and mutation mechanisms utilized for the binary variables were, respectively, the single-point crossover and the mutation rate [22]. For the continuous positions, we

Table 1
Summary of compared results.

Method and references	Parameter	Problem characteristics		Results	
				Reference	This study
Simplified equations [5]	PFD_{avg}	*SIF* (sensors *2oo3*, logic solvers *1oo2* and final elements *1oo2* with PST)		2.8E−04	2.78E−04
	STR_{avg}			1.1E−05	1.11E−05
Fault tree [9]	PFD_{avg}	*2oo3*—joint testing		1.4E−02	1.37E−02
		2oo3—staggered tests		7.3E−03	7.28E−03
Simplified equations [10]	PFD_{avg}	Highest PST interval	*1oo1*	9.40E−03	9.53E−03
			1oo2	5.86E−04	5.67E−04
			2oo2	1.91E−02	1.82E−02
			1oo3	4.77E−04	4.74E−04
			2oo3	8.05E−04	7.52E−04
			3oo3	2.86E−02	2.70E−02
		Lowest PST interval	*1oo1*	9.23E−03	9.11E−03
			1oo2	5.64E−04	5.52E−04
			1oo3	4.63E−04	4.59E−04
			2oo2	1.85E−02	1.77E−02
			2oo3	7.70E−04	7.36E−04
			3oo3	2.77E−02	2.61E−02
Fault tree [11]	PFD_{avg}	Highest test interval	*1oo1*	1.27E−01	1.30E−01
			1oo2	2.93E−02	2.68E−02
			2oo3	5.59E−02	5.09E−02
		Lowest test interval	*1oo1*	7.43E−03	7.44E−03
			1oo2	4.31E−04	4.21E−04
			2oo3	5.48E−04	5.23E−04
Fault tree [12]	PFD_{avg}		*1oo1*	4.39E−03	4.39E−03
			1oo2	8.94E−04	8.93E−04
			1oo3	8.78E−04	8.75E−04
			2oo2	7.88E−03	7.84E−03
			2oo3	9.27E−04	9.24E−04
	STR_{avg}		*1oo1*	2.50E−06	2.50E−06
			1oo2	4.65E−06	4.65E−06
			1oo3	6.80E−06	6.80E−06
			2oo2	3.65E−07	3.66E−07
			2oo3	3.95E−07	3.98E−07

used the simulated binary crossover (SBX) and polynomial mutation operator [23].

Among the four (or five) binary positions, only the first two are used directly to obtain the values of PFD_{avg}, STR_{avg} and U_{avg}. The other binary positions control the values of the continuous variables, based on the following rules:

i. Third binary position: if its value is equal to 1, the tests will not be executed in the components, so $T_{FT}=t_{max}+1$;
ii. Fourth binary position: if its value is equal to 0, the tests will be executed jointly in all components of the subsystem, so $f_{TI}=0$. If its value is equal to 1, then $1 \geq f_{TI} > N((T_D+MTTR)/(T_C - T_D - MTTR))$, to assure there is no overlap of tests or repairs of different components.
iii. Fifth binary position (only applicable to subsystems of final elements): if its value is equal to 1, the PST will not be executed, so $N_{PST}=0$.

The optimization model was established so that regardless of the number of *SIF*s in the SIS, there always exist only two objective functions. The first of them takes into account the total cost related with the SIS, composed of four portions: (i) cost of spurious failures: $\sum_{\forall\ \mathrm{SIF}} STR_{avg} C_S$, where C_S is the cost per spurious failure; (ii) costs of unavailability of production due to tests and maintenance: $\sum_{\forall\ \mathrm{SIF}} U_{avg} R$, where R is the cost of unavailability of the protected system per unit of time; (iii) cost of maintenance to repair the components of the SIS: $\sum_{\forall i} \left(\lambda_D^i + \lambda_S^i\right) C_{Man}^i$, where C_{Man}^i is the component i maintenance cost for each event; and (iv) costs of performing the tests, where C_{Test} is the cost of each complete test and C_{PST} is the cost for each partial test. The second of them is the maximum logarithmic deviation (Eq. 7) between the PFD_{avg}, resulting from the strategy applied, and the maximum acceptable PFD_{avg} for the SIL of each *SIF* ($PFD_{Avg\ max}^{SIL}$). This expression yields values lower than zero when all the *SIF*s meet the required SIL and values greater than or equal to zero when at least one *SIF* does not satisfy the required SIL.

$$\Delta_{Log}^{PFD} = \max\left\{\left[\log\left(PFD_{Avg}^{SIF}\right) - \log\left(PFD_{Avg\ max}^{SIL}\right)\right] \ \forall\ SIF\right\} \tag{7}$$

By observing the behavior of the selection mechanisms and their consistency in the various sets of solutions during the generations, we identified the opportunity to evaluate a change in the secondary selection mechanism of the NSGA-II (crowding distance), making it obtained by $1/\left|\Delta_{Log}^{PFD}\right|$ and in this way prioritizing selection of non-dominated solutions that are nearest to $\Delta_{Log}^{PFD}=0$.

After executing the genetic algorithm, which tries to minimize both the objectives, the results of all the generations were grouped, to allow identification of the final Pareto frontier.

4. Case studies

We selected three SIS configurations to assess the model presented in this paper, all with t_{max} equal to 6 years (52,560 h)

- Case i: SIS with one *SIF*; sensor elements with *2oo3* configuration; logic solvers with *1oo2* voting; and only one final element. The final element is prepared to execute the PST, but its complete testing requires interrupting production of the protected system. The required integrity level is SIL 2.
- Case ii: SIS with one *SIF*; sensor elements with *2oo3* configuration; logic solvers with *1oo2* voting; and final elements with *1oo2* configuration. The final elements are prepared to execute the PST, but their complete testing requires interrupting production of the protected system. The required integrity level is SIL 3.
- Case iii: SIS with two *SIF*s (*SIF* 1 and *SIF* 2). In this case there are two subsystems of sensor elements, both with *2oo3* voting, for each *SIF*; a single subsystem of logic solvers, with *1oo2* voting, that processes the logic of the two *SIF*s; and two subsystems of final elements, one of them with only one component (*SIF* 1) and the other with *1oo2* configuration (*SIF* 2). All the final elements are prepared to execute the PST, but their complete testing requires the interrupting production of the protected system. The required integrity levels are SIL 2 for *SIF* 1 and SIL 3 for *SIF* 2.

The reliability data of the components were obtained from a specific document [24] and are summarized in Table 2.

The genetic algorithm was executed using the NSGA-II method and the modification proposed in this article. In each case we used combinations of 50 solutions for 500 generations and 100 solutions for 250 generations, with different mutation parameters, for a total of eight different scenarios.

5. Results

In the legends of Figs. 6–11, the code NS represents the solutions obtained with the NSGA-II method, while the code NSV represents those obtained with the variation proposed here. The code P- denotes the results obtained with a population equal to 50 for 500 generations and P+ represents those obtained with population of 100 for 250 generations. The code *M*- denotes the results obtained using the lowest mutation rates and *M*+ denotes those obtained with the highest mutation rates.

For each case we present the Pareto frontiers obtained with the eight different scenarios. Fig. 6 presents the final Pareto frontiers obtained for case i, Fig. 7 for case ii and Fig. 8 for case iii. These Pareto frontiers are exhibited in the interval $[-1,+1]$ of Δ_{Log}^{PFD}, along with detail of the $[-0.3,+0.3]$ region of Δ_{Log}^{PFD}, since this is the region of greatest interest, with respect to satisfaction of the required integrity levels.

We also present the evolutions over the generations of the lowest total cost values obtained from the solutions with $\Delta_{Log}^{PFD} < 0$, i.e., those that satisfy the required integrity levels (SIL) for all the *SIF*s. These solutions ($min\ C_{TOT} | \Delta_{Log}^{PFD} < 0$) are called here "best solution obtained". Fig. 9 depicts this evolution for case i, Fig. 10 for case ii and Fig. 11 for case iii.

Next we present the parameters of the lowest cost strategies with $\Delta_{Log}^{PFD} < 0$ obtained for each case studied, their $PFD(t)$, $STR(t)$ and $U(t)$ curves, as well as the values of PFD_{avg}, STR_{avg} and U_{avg}.

Table 2
Set of data used in formulating the case studies.

Parameter	Sensor elements	Logic solvers	Final elements
λ_{DD} (h^{-1})	0.5×10^{-6}	0.9×10^{-6}	1.1×10^{-6}
λ_{DU} (h^{-1})	0.3×10^{-6}	0.1×10^{-6}	2.9×10^{-6}
λ_{SD} (h^{-1})	0.2×10^{-6}	0.0	0.0
λ_{SU} (h^{-1})	0.3×10^{-6}	1.0×10^{-6}	4.6×10^{-6}
P_{TIF}	5.0×10^{-4}	5.0×10^{-6}	1.0×10^{-4}
β	0.03	0.02	0.02
$MTTR$ (h)	2.0	6.0	4.0
C_{Test} ($)	800.00	1600.00	1600.00
C_{Man} ($)	500.00	3000.00	3500.00
T_D (h)	1.0	2.0	2.0
θ_{pst}	–	–	0.65
C_{PST} ($)	–	–	1600.00
P_{TIS}	0.01	0.01	0.01
	Parameters of the protected system		
$MTTSS$ (h)	48		
C_S ($)	2,000,000.00		
R ($/h)	20833.33		

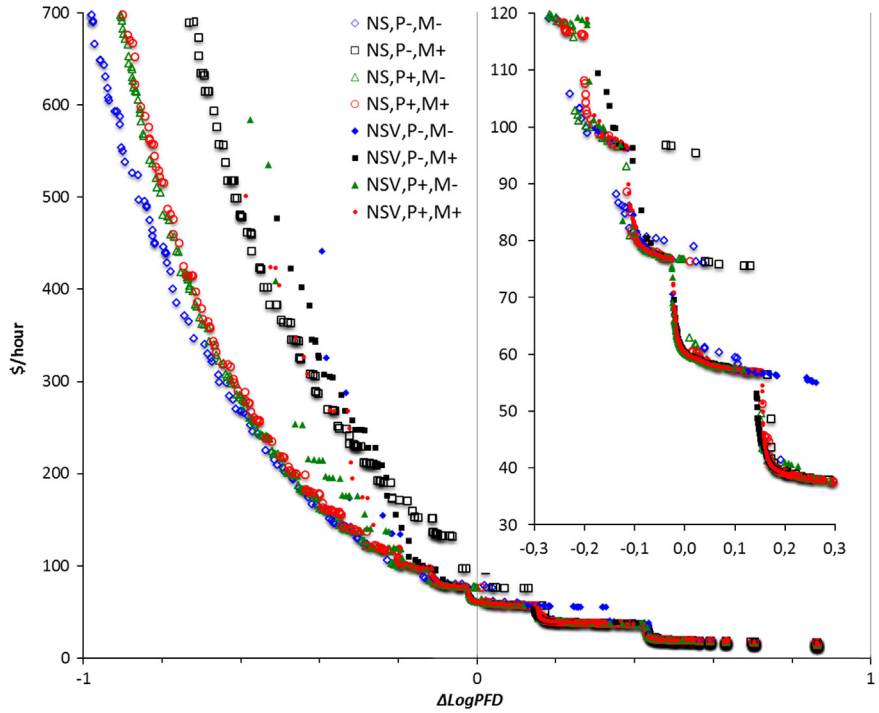


Fig. 6. Pareto frontiers for case i.

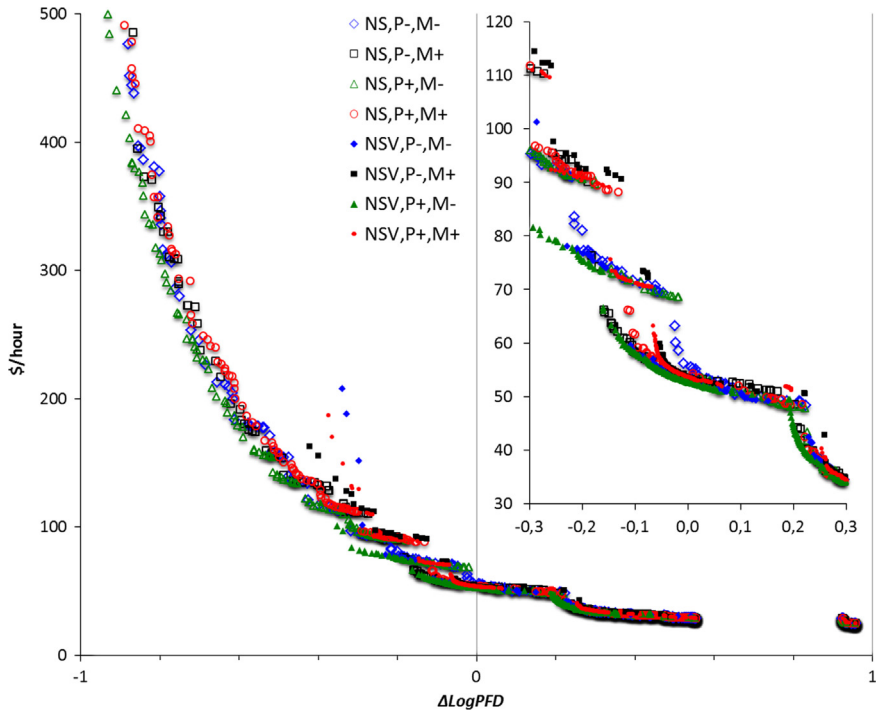


Fig. 7. Pareto frontiers for case ii.

Table 3 presents the parameters that define the best solution obtained for case i. This strategy was obtained by using the modified NSGA-II model, executed for 250 generations with 100 solutions and the highest mutation rates.

Fig. 12 presents the $PFD(t)$, $STR(t)$ and $U(t)$ curves for the solution of Table 3. The values of PFD_{avg}, STR_{avg} and U_{avg} were, respectively, 1.00×10^{-2}, $6.90 \times 10^{-6}$and 0.21%. The total cost obtained with the referred strategy was \$60.33/h.

Table 4 presents the parameters that define the best solution obtained for case ii. This strategy was obtained with the use of the modified NSGA-II model, executed for 500 generations with 50 solutions and with the lowest mutation rates.

Fig. 13 presents the $PFD(t)$, $STR(t)$ and $U(t)$ curves for the solution of Table 4. The values of PFD_{avg}, STR_{avg} and U_{avg} were, respectively, 9.99×10^{-4}, 1.18×10^{-5} and 0.14%. The total cost obtained with the referred strategy was \$52.60/h.

Table 5 presents the parameters that define the best solution obtained for case iii. This strategy was obtained with use of the modified NSGA-II model, executed for 250 generations with 100 solutions and with the highest mutation rates. The first values

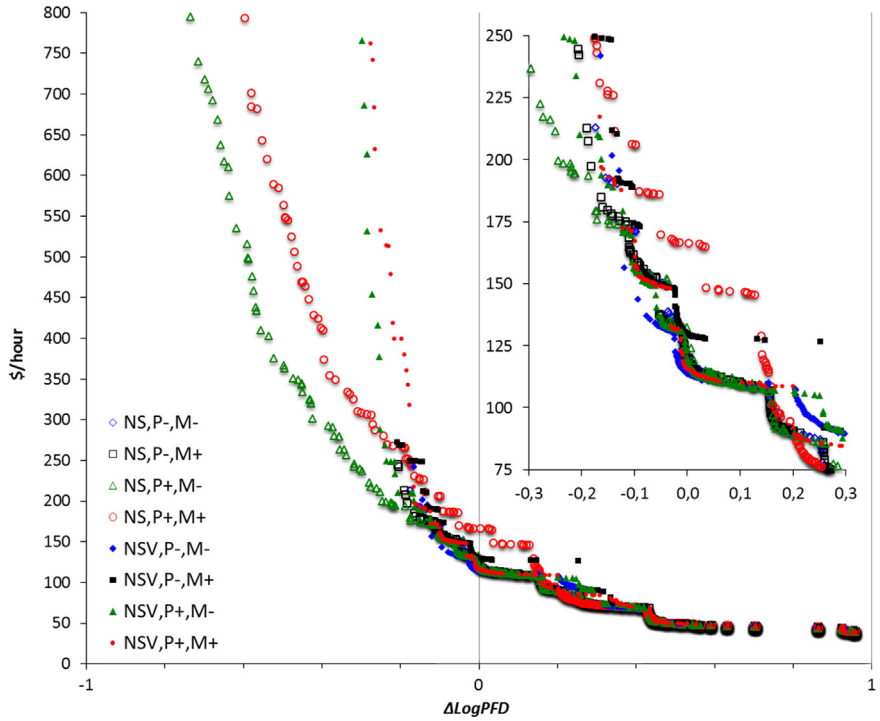


Fig. 8. Pareto frontiers for case iii.

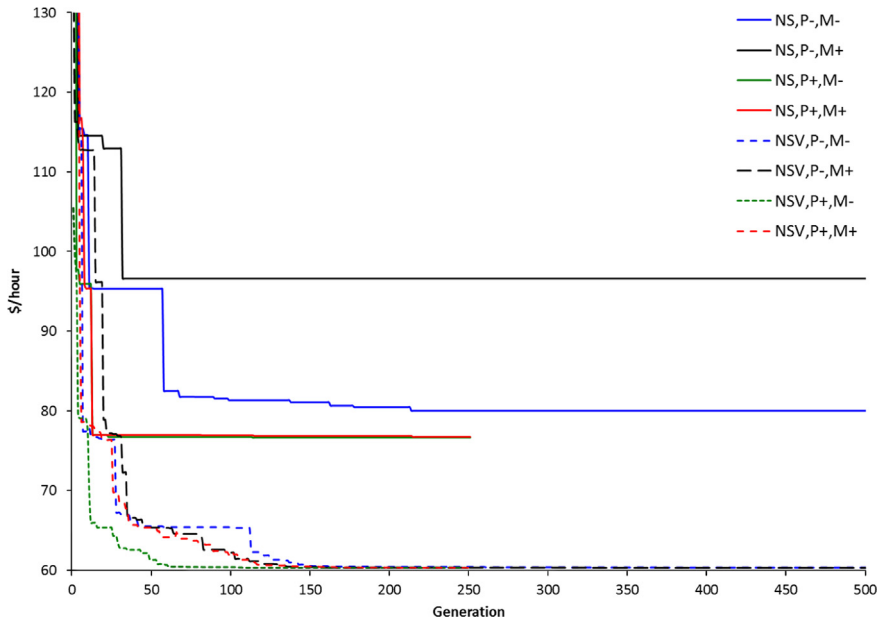


Fig. 9. Evolution of the cost result of the best solution obtained for case i.

presented refer to the subsystems of *SIF* 1 and the second to the subsystems of *SIF* 2.

Fig. 14 presents the $PFD(t)$, $STR(t)$ and $U(t)$ curves for the solution of Table 5, for both *SIF*s. The values of PFD_{avg}, STR_{avg} and U_{avg} were, respectively, 9.96×10^{-3}, 6.87×10^{-6} and 0.21% for *SIF* 1 and 9.82×10^{-4}, 1.20×10^{-5} and 0.14% for *SIF* 2. The total cost obtained with the referred strategy was \$115.85/h.

6. Final considerations

Safety instrumented systems have fundamental importance to assure the reliability, integrity and safety of industrial processes in a wide range of areas. The results presented here directly expose the conflict between costs and safety level, and demonstrate the importance of choosing a suitable operation and testing strategy. This strategy should seek a compromise between those aspects, assuring the required SIL while minimizing the associated costs. We also highlight the importance of assessment of the strategies in the design stage of the SIS, so that adequate integrated support can be provided for its operation, along with a configuration that minimizes the life cycle cost, considering the required SIL.

In this study, we developed models to represent the probabilistic behavior of these systems, and then integrated these models with multi-objective optimization algorithms, based on meta-heuristics. The aim of this approach is to allow identifying

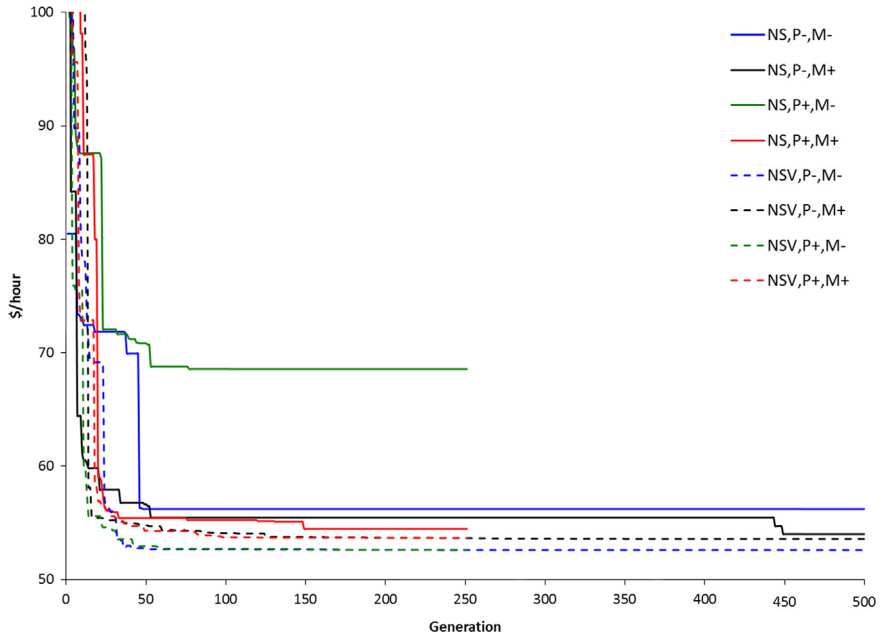


Fig. 10. Evolution of the cost result of the best solution obtained for case ii.

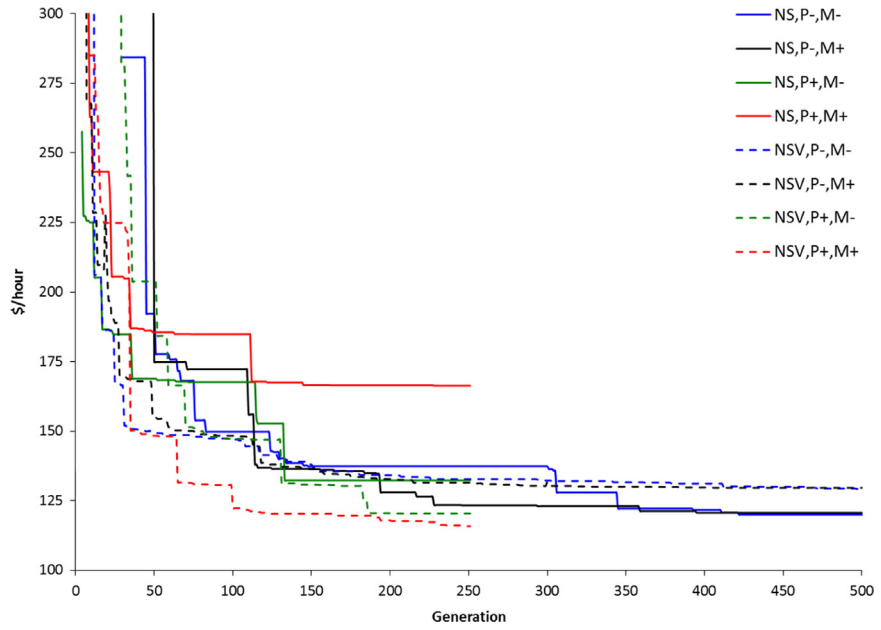


Fig. 11. Evolution of the cost result of the best solution obtained for case iii.

Table 3
Parameters of the operation and testing strategy for case i.

Parameter	Sensor elements	Logic solvers	Final elements
Operation and testing mode	*M*4	*M*3	*M*1
T_{FT} (h)	29,108	–	16,998
T_C (h)	52,385	–	17,823
f_{TI}	0	–	–
N_{PST}	–	–	26

operation and testing strategies to maximize productive efficiency and minimize costs related to guaranteeing the integrity of the active protections that operate under low demand in industries with continuous processing. The method developed was applied to three cases, allowing identification of various workable strategies, seeking a compromise between the objectives by assuring the required SIL while minimizing the related costs. We did not evaluate the uncertainties related to the reliability and maintainability parameters, so these should be quantified in future studies.

Some strategies identified, among them those presented in Tables 2 and 4, make use of a large number of partial tests of the final elements, requiring their execution at intervals on the order of 30 days. That fact highlights the importance of including factors in the probabilistic model related to the deterioration or wear that some types of components can undergo because of successive activation, be they total or partial. The model presented here disregards these factors.

When comparing the Pareto frontiers obtained in the three cases studied by the NSGA-II method [21] with the frontiers obtained by the modified NSGA-II method, presented in Figs. 6–8, it can be seen that in the regions with $\Delta_{Log}^{PFD} \approx 0$, the modified NSGA-II method permitted identifying more efficient operation

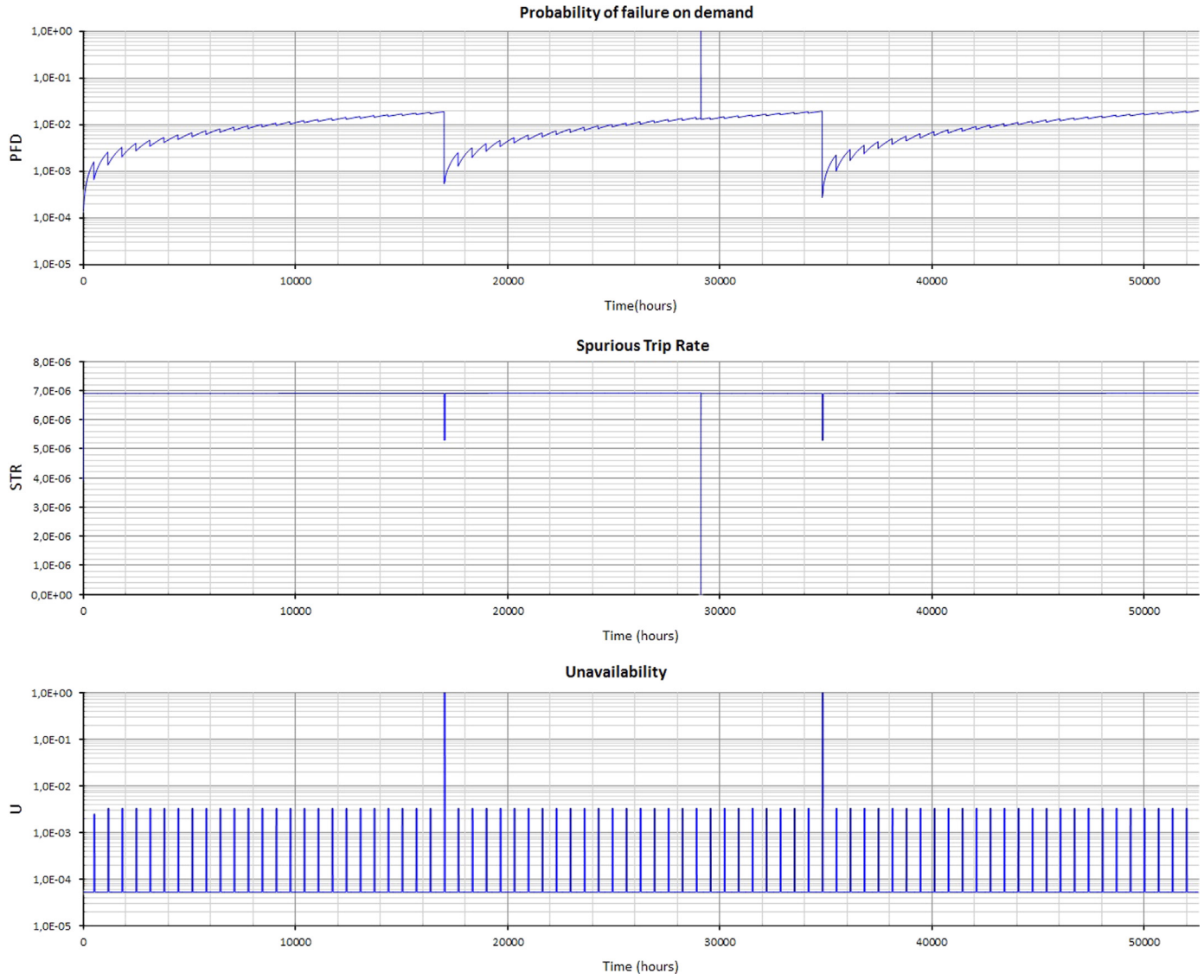


Fig. 12. $PFD(t)$, $STR(t)$ and $U(t)$ curves for case i.

Table 4
Parameters of the operation and testing strategy for case ii.

Parameter	Sensor elements	Logic solvers	Final elements
Operation and testing mode	*M*4	*M*3	*M*1
T_{FT} (h)	13,367	–	25,261
T_C (h)	14,936	–	34,226
f_{TI}	0	–	0
N_{PST}	–	–	4

and testing strategies in terms of harmonization of the objectives of the problem. In turn, when evaluating the regions farther from $\Delta_{Log}^{PFD} \approx 0$, the original NSGA-II method enabled identifying more efficient solutions. That pattern can be explained because with use of the NSGA-II method, the selection by crowding distance (Cd) prioritizes maintenance over the generations with more dispersed solutions in the Pareto frontier, causing greater diversity of solutions along the entire amplitude of the domain of each of the objectives. However, this selection mechanism in some cases disregards entire solution regions that present good results for both objectives, because of the fact they are in regions that have a greater degree of agglomeration. On the other hand, the modified NSGA-II method, as proposed and assessed in this study, uses as selection indicator a Cd value that is considered to be better the nearer the solution is to $\Delta_{Log}^{PFD} = 0$. This mechanism does not prioritize the selection of dispersed solutions with variation of the objectives, so it does not produce such ample Pareto frontiers as those obtained by the NSGA-II method. However, this modification prevents good solutions that are near $\Delta_{Log}^{PFD} = 0$ from being discarded over the generations, resulting in better identification of operation and testing strategies with significantly lower costs in this region.

With respect to the influence of the mutation rate and number of solutions and generations, the variation of the cost results of the best solution obtained for the cases tested was significantly lower with use of the modified NSGA-II model than with the NSGA-II model. In case i (Fig. 9), with the NSGA-II model this variation was 26.0%, while with the modified NSGA-II model it was only 0.04%. In case ii (Fig. 10), these values were 26.9% and 2.0% and in case iii (Fig. 11) they were 38.6% and 11.9%, respectively. These results indicate less influence of the parameters that govern the optimization model on the results obtained with use of the modified NSGA-II algorithm in comparison with the original NSGA-II algorithm. Based on this evidence, together with the fact the modified NSGA-II method allowed identifying operation and testing strategies with significantly lower costs, it can be concluded that the

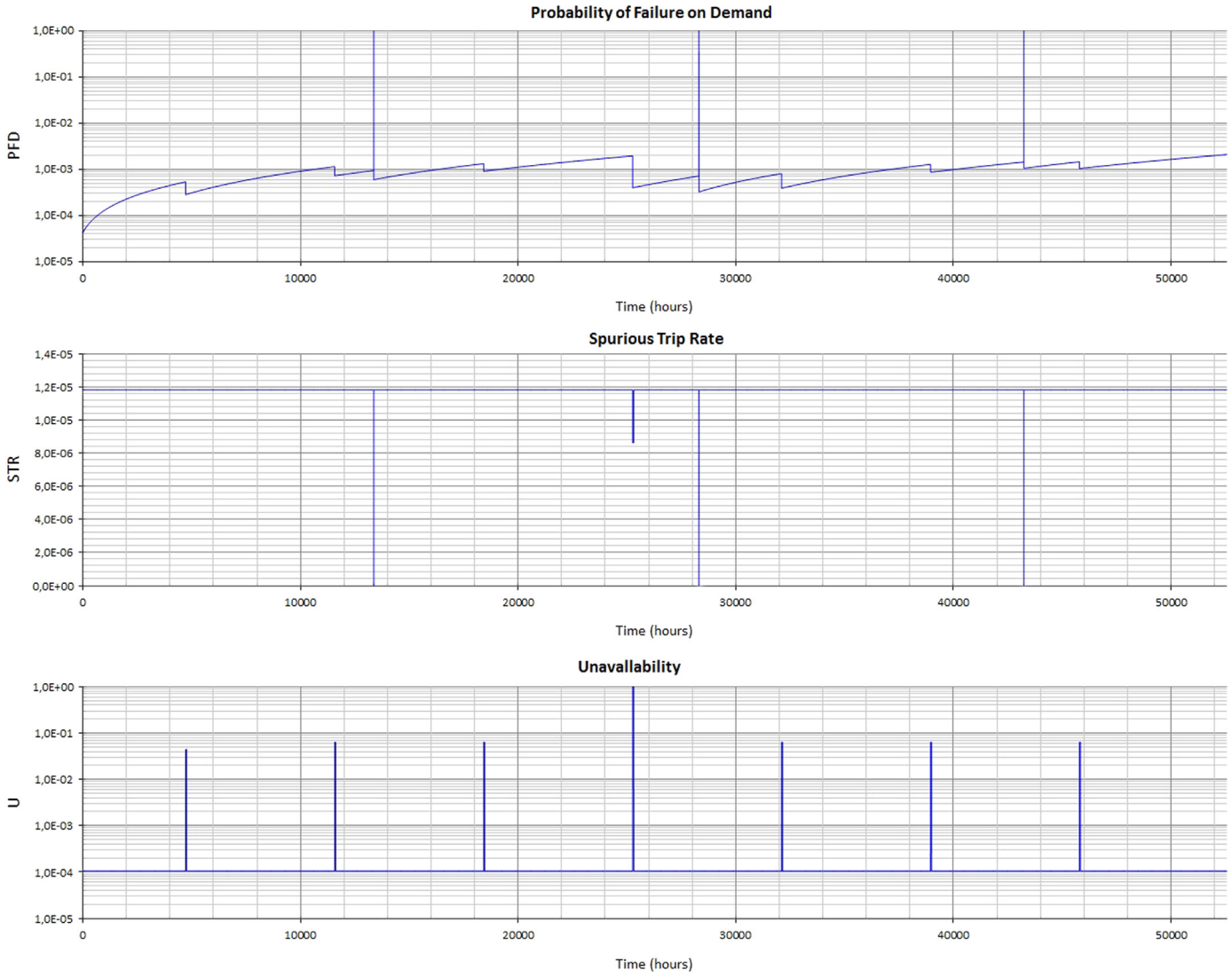


Fig. 13. $PFD(t)$, $STR(t)$ and $U(t)$ curves for case ii.

Table 5
Parameters of the operation and testing strategy for case iii.

Parameter	Sensor elements	Logic solvers	Final elements
Operation and testing mode	*M*4|*M*3	*M*3	*M*1|*M*1
T_{FT} (h)	26,824|21,730	–	17,853|26,000
T_C (h)	52,560|52,560	–	18,147|32,680
f_{TI}	0|0.362	–	–|0
N_{PST}	–	–	27|5

modified NSGA-II method allows identifying more efficient operation and testing strategies in the region of $\Delta_{Log}^{PFD} \approx 0$, with less dependence on the optimization parameters.

The advantage of using staggering tests as a mechanism to reduce the PFD_{avg}, as exposed in [2,9], was not evidenced in this study when considering the formulation of STR_{avg} that considers a probability of failure induced by the test (P_{TIS}) with values on the order of 1%. We carried out some tests considering P_{TIS} on the order of 0.1%, when solutions with $f_{TI} > 0$ started to appear more frequently among the solutions of the final Pareto frontier (data not shown). These results demonstrate the importance of training, planning and control of the performance of the tests in SISs, to assure very low failure levels during their execution, thus allowing the use of scaled tests to obtain a greater reduction of the PFD_{avg} value.

With respect to the different operation and testing modes available for selection and use, it is important to note that each of these modes can be enabled or disabled for selection by the optimization algorithm in each of the subsystems of the SIS under analysis. That fact is extremely important since in some subsystems, such as the subsystem of final elements, the applicability of mode *M*3, for *1ooN* systems, or mode *M*4 for the other voting logic systems, depends on physical configuration of the system in the field. In the majority of these cases it is necessary to have an alternative path (bypass) for outflow of production while these components are tested. This configuration is not common in industrial installations. Nevertheless, the study presented in [15] adopted two possible modes of conducting these tests, equivalent to *M*2 and *M*3. In that study, the authors assumed that the testing mode is a consequence of the redundancy configuration of the components of each subsystem of the *SIF*, so the tests realized in systems with *1ooN* configuration used the closed test mode (*M*3) and the other systems with *MooN*, with $M > 1$, used the open test mode (*M*2). In the case of subsystems of final elements with *1ooN* configuration and without bypass in the industrial installation (a common configuration for these subsystems), leads to the

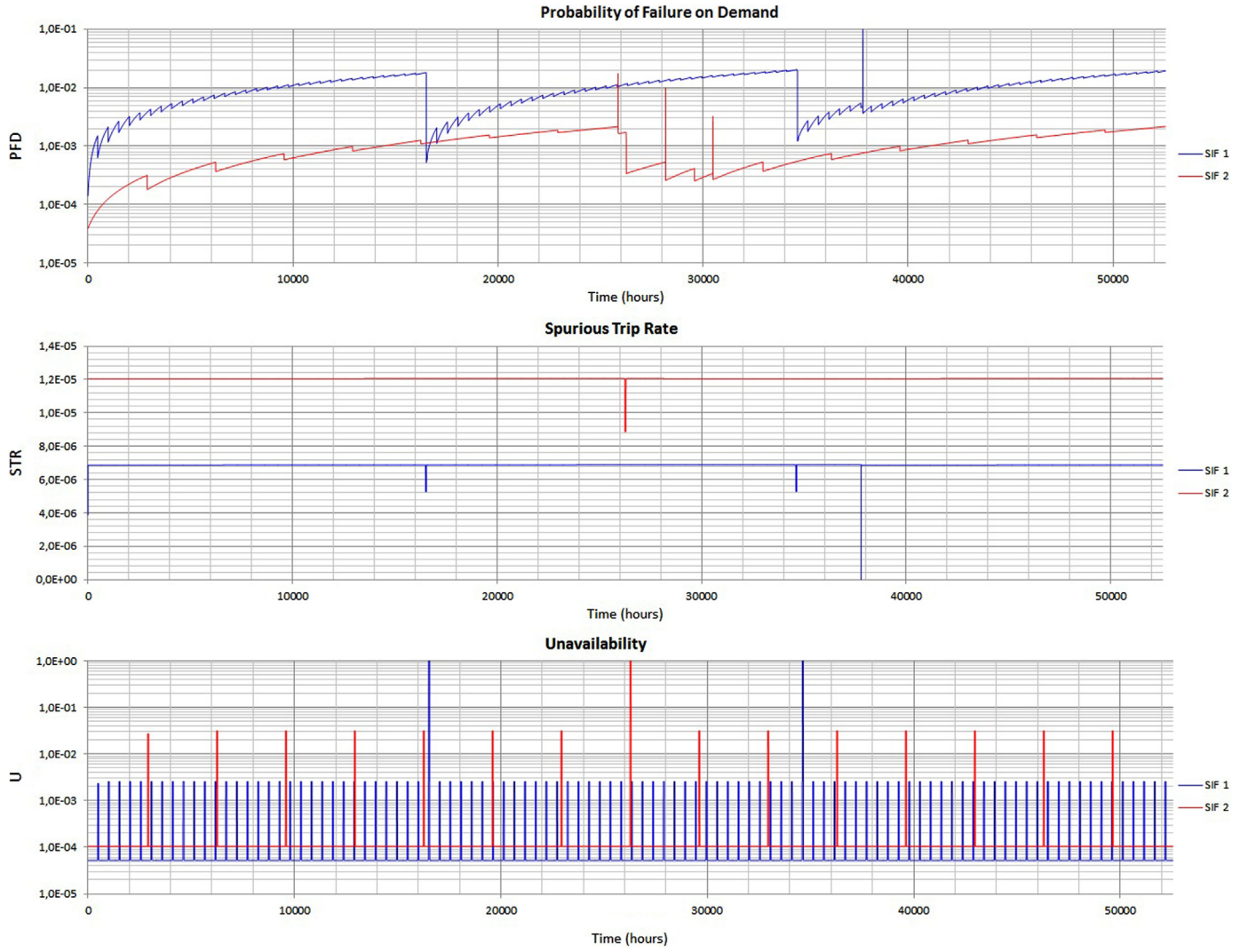


Fig. 14. *PFD*(t), *STR*(t) and *U*(t) curves for case iii.

establishment of an operation and testing strategy that is unworkable from a practical standpoint.

Regarding identification of operation and testing strategies, analysis of case iii indicates that it is basically the union of case i and case ii, with the reservation that in case iii, the logic solver subsystem is common to the two *SIF*s. Considering that in cases i and ii it was not necessary to carry out tests of the logic solver subsystem, we can conclude that the strategy for case iii could be obtained by joining the strategies for cases i and ii, with the result being the sum of the C_{TOT} values of cases i and ii, excluding the maintenance costs of one logic solver subsystem. Therefore, for case iii we expected a C_{TOT} result of $ 112.92/h. However, the real value obtained was $ 115.85/h, a deviation of 2.6%. That fact shows that the search for individual solutions for each *SIF* can bring better results than searching for solutions for the whole SIS, since the search space is smaller. However, if the subsystem shared among various *SIF*s is subject to different testing strategies in each of the cases, the individual analysis of each *SIF* to compose the strategy for the SIS would not be applicable, as in the case presented, so joint formulation for more than one *SIF* is important.

It should be noted that since in general the final element subsystems are those that are least reliable, and consequently require more frequent testing, and considering that in most situations their complete testing requires shutting down the productive process, it is important to use these opportunities to carry out the tests in advance of their normal schedule. These opportunities generally arise from shutdowns due to equipment failure or low market demand for the product made. Once the complete tests in the *SIF* are realized on such opportunities, the optimization model described here can be carried out again, considering a t_{max} equal to the remaining operation time of the protected system until the next scheduled testing window (general shutdown for maintenance). This practice reduces the impacts of the tests on the life cycle costs of the installation, improving business performance in terms or risk and cost.

The main limitations of the method presented here are: (i) the model is only applicable to SISs in low-demand operation mode; (ii) the model represents homogeneous components in each of the SIS subsystems; (iii) the behavior of the time to failure of each component is described in the exponential distribution; (iv) the model is only applicable to *SIF*s that can be represented by a subsystem of sensors, a subsystem of logic solvers and a subsystem of final elements, so its availability depends on the availability of all of them; (v) the model is only able to identify strategies with constant intervals between tests; (vi) the probabilistic model does not represent the degradation of the components due to execution of the tests, whether complete or partial; and (vii) the proposed model does not assess the impact of uncertainty about the reliability parameters, together with the defined strategies, on the results.

References

[1] Center for Chemical Process Safety. Layer of protection analysis–simplified process risk assessment. New York: CCPS/AICHE; 2001.
[2] Rausand M, Høyland A. System reliability theory: models, statistical methods, and application. 2nd ed. New Jersey: Wiley; 2004.
[3] International electrotechnical commission. IEC 61511. Functional safety—safety instrumented systems for the process industry sector. Switzerland, Part 1–3; 2004.
[4] International electrotechnical commission. IEC 61508. Functional safety of electrical/electronic/programmable electronic safety-related systems. Switzerland, Part 1–7; 2010.
[5] SINTEF. Reliability prediction method for safety instrumented systems – PDS method handbook. 2010 ed.. . Trondheim: SINTEF; 2009.
[6] Torres-Echeverría AC, Martorell S, Thompson HA. Modeling and optimization of proof testing policies for safety instrumented systems. Reliab Eng Syst Saf 2009;94:838–54.
[7] Lundteigen MA, Rausand M. Spurious activation of safety instrumented systems in the oil and gas industry: basic concepts and formulas. Reliab Eng Syst Saf 2008;93:1208–17.
[8] HSE. Principles for proof testing of safety instrumented systems in the chemical industry. Health and Safety Executive; 2002, 〈www.hse.gov.uk/reserch/crr_pdf/2002/crr02428.pdf ISBN 0 7176 2346 7〉.
[9] Signoret JP. High integrity protection systems (HIPS): methods and tools for efficient safety integrity levels (SIL) analysis and calculations. SPE Proj Facil Constr 2008;6:1–6.
[10] Oliveira, LF. Evaluating the PFD of instrumented safety systems with partial stroke testing. In: Proceedings of the international conference on probabilistic safety assessment and management; 2008.
[11] Brissaud, F; Oliveira, LF. Average probability of a dangerous failure on demand: different modelling methods, similar results. In: Proceedings of the European safety and reliability conference; 2012.
[12] Innal F, Dutuit Y, Chebila M. Safety and operational integrity evaluation and design optimization of safety instrumented systems. Reliab Eng Syst Saf 2015;134:32–50.
[13] Torres-Echeverría AC, Martorell S, Thompson HA. Multi-objective optimization of design and testing of safety instrumented systems with MooN voting architectures using genetic algorithm. Reliab Eng Syst Saf 2012;106:45–60.
[14] Alijagic, E; Dang, VN. Design of safety functions form complex systems: treating multiple hazards and shared components. In: Proceedings of the European safety and reliability conference; 2012.
[15] Torres-Echeverría AC, Martorell S, Thompson HA. Modeling safety instrumented systems with MooN architectures addressing system reconfiguration for testing. Reliab Eng Syst Saf 2011;96:545–63.
[16] Lundteigen MA, Rausand M. Partial stroke testing process shutdown valves: how to determine the test coverage. J Loss Prev Process 2008;21:579–88.
[17] Lundteigen, MA; Rausand, M. The effect of partial stroke testing on the reliability of safety valves. In: Proceedings of the European safety and reliability conference; 2007.
[18] Fleming KN. Reliability model for common mode failures in redundant safety systems. San Diego: General Atomic Company; 1975 Report GA-A13284.
[19] Marseguerra M, Zio E, Martorell S. Basics of genetic algorithms optimization for RAMS applications. Reliab Eng Syst Saf 2006;91:977–91.
[20] Konak A, Coit DW, Smith AE. Multi-objective optimization using genetic algorithms: a tutorial. Reliab Eng Syst Saf 2006;91:992–1007.
[21] Deb K, Pratap A, Agarwal S, Meyarivan T. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Trans Evol Comput 2002;6:182–97.
[22] Xinjie Y, Mitsuo G. Introduction to evolutionary algorithms. London: Springer; 2010.
[23] Deb K, Agrawal R. Simulated binary crossover for continuous search space. Complex Syst 1995;9:115–48.
[24] SINTEF. Reliability data for safety instrumented systems: PDS data handbook. 2010 ed. Trondheim: SINTEF; 2009.